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Summary

I develop identity and access management (IGA/IAM) solutions in ways that people can understand and trust. I work on data protection throughout the identity lifecycle in advanced access control systems. I have designed and developed techniques for authentication anomaly detection. Our work has been recognized multiple times by the leading conference in the field, ACM SACMAT, and by the Microsoft Bounty Program.

Education

PhD in Information Security, Eindhoven University of Technology (TU/e) 2022 – 2026

Supervisor: Dr. Nicola Zannone

Area of expertise: data protection through advancing access control in complex environments

- *Anomaly detection to prevent access privilege misuse.* Built profiles of user authentication behavior for Microsoft Entra ID (formerly Azure Active Directory) and healthcare environments. Designed framework for real-time monitoring of access requests to detect deviations indicating potential privilege misuse. Applied machine learning with advanced domain-informed feature engineering and techniques for handling sparse historical access events.
- *Enhancing understandability of access control policies and decisions.* Introduced the concept of “Explainable Access Control” to address the need for better understanding access policies and decisions throughout identity lifecycle stages and across different stakeholder perspectives. Developed two complementary frameworks as implementations of this vision: a tool based on causal reasoning to identify policy elements responsible for access outcomes, and a visualization framework for explaining XACML policy evaluation, implemented in SAFAX.
- *Automating access decision making in collaborative settings.* Automated access decision making using game theory to account for multiple stakeholders’ preferences and resolve access conflicts. Investigated collaborative access control in online social networks and evaluated deep learning techniques to edit images while preserving image fidelity and users’ privacy expectations.

Skills

IAM Governance: Identity lifecycle, access reviews, least-privilege, separation (segregation) of duties

Access Control Models: ABAC, RBAC, PBAC

Access Control Approaches: Risk-based access control, adaptive authentication, conditional access control

Programming: Python (IEEE certified), R, C++

Language: English (TOEFL iBT 113/120), Dutch (A2 level)

Experiences

Modeling cloud authentication access behavior through user profiling for Microsoft Entra-ID 2026

Developed user profiles to model authentication behavior on Microsoft Entra ID following user and entity behavior analytics (UEBA) techniques. Applied machine learning methods (including clustering and classification) to capture baseline authentication behaviors, in order to detect anomalies in sign-in patterns or inform conditional access policies. Context-aware feature engineering was employed, while handling challenges related to limited historical data. Results were validated through sessions with experts at KEMBIT, a Microsoft partner in the Netherlands, confirming interpretable and informative insights into users’ authentication behavior. *Publication: Behavior-Driven User Profiling: A Case Study on Microsoft Entra ID Sign-In, Journal of Computer Security (Sage Journals, 2026)*

Supporting comprehension of access control decisions through causality analysis 2026

Proposed a framework to support explaining access control decisions by identifying the policy elements causing the outcomes. Introduced metrics to quantify the influence of these elements on access decisions. This work is

part of the broader concept of explainable access control, which enables access policy analysis and helping stakeholders detect unintended behavior and guide systematic policy refinement.

Publications: *A Causal Framework for Explainable Access Control*, ACM Symposium on Access Control Models and Technologies (SACMAT), Waterloo, Canada, 2026. and *Explaining Access Control Through Causal Reasoning*, International Conference on Information and Communications Security (ICICS), Japan, Fukui, 2026.

Personalized privacy control using deep learning techniques for image localization and inpainting in online social networks 2025

Proposed a privacy model for online social networks that enables users to conceal sensitive content in shared images based on personalized access policies, while preserving image fidelity. Implemented a pipeline that performs localization using deep learning models such as MaskDino and SAM, and inpainting with LaMa and SD. Validated with real users to confirm the solution preserved image quality while meeting privacy expectations.

Publication: *Personalized Privacy in OSNs: Evaluating Deep Learning Models for Context-Aware Image Editing*, Intelligent Systems with Applications (Elsevier), 2025

Foundations for explainable access control 2025

Introduced for the first time “Explainable Access Control” to address the needs for better understanding access policies and decisions. Developed a systematic framework for explaining access decisions across key dimensions: what gets explained, to whom (administrators vs. stakeholder vs. auditors), and when in the access lifecycle.

Publication: *Towards Explainable Access Control*, ACM Symposium on Access Control Models and Technologies (SACMAT), New York, USA, 2025 (**Best Paper Award**)

Intrusion detection opportunities in Microsoft Purview Unified audit logs 2024

Analyzed detection capabilities of Microsoft Purview Unified audit logs for attacks on Exchange Online and SharePoint Online. Proposed an extended ATT&CK Matrix for Office 365 and log-based features for use in anomaly-based intrusion detection.

Publication: *Opportunities for Intrusion Detection in Cloud Productivity Environments*, TU Eindhoven, 2024 (**Microsoft Bounty Program Awarded**)

Decision making for collaborative access control through game-theoretic modeling 2024

Proposed a framework for access control in collaborative settings where multiple stakeholders manage shared resources. Developed a game-theoretic model that accounts for the privacy preferences of multiple parties and automates decision-making. The model ensures acceptability and optimality of collective access decisions without requiring user intervention. Explored various strategic environments (cooperative vs. non-cooperative bargaining) to represent distinct decision dynamics.

Publication: *A Bargaining-Game Framework for Multi-Party Access Control*, ACM Symposium on Access Control Models and Technologies (SACMAT), San Antonio, USA, 2024 (**Runner-up Best Paper**)

Visualizing XACML policies and their evaluations to support comprehension 2024

Created visualization framework to support understanding of XACML (an XML-based standard markup language) policy evaluations and their contribution to access decisions. Enabled both technical and non-technical stakeholders to trace how decisions are derived and improved transparency and trust. Integrated the framework with SAFAX.

Publication: *Enhancing the Comprehension of XACML Policies*, ACM Symposium on Access Control Models and Technologies (SACMAT), San Antonio, USA, 2024

Anomaly detection for preventing privilege misuse in access control 2023

Designed a real-time identity threat detection and response (ITDR) framework for monitoring access requests throughout the identity lifecycle in healthcare. Validated against hospital access data including the Break the Glass (BTG) protocol for emergency patient record access, achieving a low false positive rate. Detected privilege misuse patterns during provisioning and mover events that bypass standard authorization controls and escape detection by traditional compliance auditing.

Publication: *Mitigating Privilege Misuse in Access Control through Anomaly Detection*, International Conference on Availability, Reliability and Security (ARES), Benevento, Italy, 2023

Professional Competencies

- Gather/explain access requirements/framework from/to non-technical stakeholders
- Communicate authentication anomaly findings to security and compliance teams
- Maintain expertise through continuous learning

References

- [Nicola Zannone](mailto:n.zannone@tue.nl), *n.zannone@tue.nl*